Zeyu (Alban) LI

Materials Scientist | Polymers & Hydrogels | 626-826-5327 | zl788@cornell.edu | zeyuli.net | Available May 2026

Materials scientist specializing in polymer and biopolymer systems--hydrogel formulation, microparticle encapsulation, and functional surface engineering. Expertise in processing-structure-property relationships across diverse crosslinking chemistries, with full-cycle R&D from formulation and fabrication through characterization (rheometry, SEM, tensile testing) to field-scale validation. Developed a PEG hydrogel composite with 100-day stability (Nature under review), PLGA microparticle tracers deployed across 11 km2 for DoD (first-author, ES&T), and durable perfluoropolymer superhydrophobic surfaces (The Innovation IF 33.2). Co-authored four publications and co-invented two patents.

SKILLS

Materials Characterization: SEM; DLS; EDS; Optical microscopy; UV-Vis spectroscopy; Fluorescence microscopy; Oscillatory rheometry; Tensile testing (Instron); Contact-angle measurement.

Fabrication & Processing: Electrospinning; 3D printing (DIW, SLA, FDM); Thermal imprinting; Double-emulsion microencapsulation; Hydrogel formulation; UV photocrosslinking; Spin coating; Mold casting; Photolithography; Two-photon lithography; Etching; Nanopatterning; Freeze-drying; Roll-to-roll processing

Programming & Data/Modeling: Python; Data analysis & visualization (pandas, NumPy, seaborn); Machine learning (TensorFlow, PyTorch - working knowledge); Java; AI tools (computational workflows, LLM-assisted data analysis)

2D/3D Design & Visualization: CAD; SketchUp; Autodesk 3ds Max; Blender; KeyShot; Adobe Illustrator, Photoshop, InDesign; Bootstrap Studio; Scientific figure production; Video/audio editing.

Molecular & Bioengineering: PCR; qPCR; IVT; Electrophoresis (PAGE, agarose); DNA/RNA extraction & purification; LSPR-based binding assays; Cleanroom protocols; Chromatography; Enzymatic assays and workflows.

EDUCATION

Cornell University, Ithaca, NY

Biological and Environmental Engineering

Concentration: *Materials Science*, *Biological Engineering*, *Bioenvironmental Engineering*

Ph.D. -- DNA Materials Lab, Advisor: Dan Luo | Expected May 2026

M.S. | Aug 2024

M.Eng. | May 2020

Hong Kong Baptist University, Kowloon, Hong Kong

Chemistry (Major) and Computer Science (Minor)

B.Sc. (Hons) -- Microfabrication & Surface Materials Lab, Supervisor: Kangning Ren | Nov 2019

EXPERIENCE

Graduate Research Assistant | Ithaca, NY

DNA Materials Lab, Cornell University | Feb 2021 - Present

- **Developed biodegradable PLGA microparticles encapsulating synthetic DNA via double-emulsion solvent evaporation** as hydrological tracers; SEM-characterized size distribution matched natural eDNA particle range. Produced gram-scale batches and led a DoD-funded field deployment across 11 km2 of lake, achieving qPCR detection 7 km downstream from less than 1 mg release. (first-author, ES&T, 2025).
- **Engineered a DNA-functionalized PEG hydrogel-mesh composite (NucleoMesh) for continuous biomanufacturing:** UV-cured organogel on polyester mesh scaffold with covalent DNA immobilization (NHS-amine bioconjugation); systematically varied monomer concentration, crosslinker ratio, and template loading, elucidating non-monotonic process-structure-property relationships. Characterized by SEM, fluorescence microscopy, rheometry, and UV-Vis spectrophotometry; validated 100-day material stability under continuous flow. (2nd author, Nature under review.)

- **Developed biomass DNA hydrogels with three crosslinking chemistries--thermal, photochemical (UV-reversible psoralen), and ionic (Al3+)**--producing stimuli-responsive networks spanning thermoreversible to irreversible from a single biopolymer feedstock. Mapped processing-structure-property relationships across 6 formulation variables; characterized by oscillatory rheometry and Instron tensile testing. Demonstrated extrusion-based 3D printing and vascular self-healing in ceramic tiles. (co-first-author, ACADIA, 2024).
- **Invented an automated DNA feedstock extraction platform** inspired by espresso machines, delivering ~200 mg DNA per pod and reducing production cost by ~91%.
- **Mentored 3 undergraduate researchers and co-instructed Design and Analysis of Biomaterials** (~50% of content, ~45 students): designed a 6-lecture AI tooling module and organized a 40-student Biomaterials Hackathon with structured deliverables and oral defense format. Guest lecturer on computational workflows for 2 additional courses; teaching assistant across 7 engineering courses total.

Undergraduate Research Assistant & Senior Research Assistant | Kowloon, Hong Kong
Hong Kong Baptist University | Jun 2017 - Aug 2019 & Oct 2020 - Jan 2021

- **Engineered durable monolithic perfluoropolymer superhydrophobic surfaces** by replicating hierarchical nano-micro topologies via thermal imprinting with master molds from photolithography, two-photon lithography, and etching; validated superwettability and ultradurability across self-cleaning, chemical, thermal, and mechanical stress testing. (3rd-author paper, The Innovation IF 33.2; US Patent 11,839,998).
- **Led an experimental study on static-charge anti-icing,** demonstrating ~25% reduction in droplet contact time. Received the Best Undergraduate Thesis Award.

Research Exchange Trainee | Atlanta, GA
Georgia State University, Molecular Basis of Disease Program | Jun 2018 - Aug 2018

- **Characterized protein-DNA binding kinetics** using LSPR (localized surface plasmon resonance) assays. Measured how transcription factor PU.1 interacts with DNA, providing insights into sequence-specific binding affinities in real time. Received the Best Poster Presentation Award for this work.

PUBLICATIONS

- Wang, D., ***Li, Z.***, Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." Nature, under review (2026).
- ***Li, Z.,*** Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling" Environmental Science & Technology (IF = 12.4), (2025). DOI
- He, C.*, ***Li, Z.*,*** Wang, L. X., et al. "*PolyTile 4.0: Self-healing Ceramic Tiles*" ACADIA (2024). DOI
- Li, W., Chan, C. W., ***Li, Z.***, et al. "*All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability.*" The Innovation (IF = 33.2), 4(2), 100299, (2023). DOI

PATENTS

- Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., ***Li, Z***., Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998, (2023).
- Li, Q., ***Li, Z.,*** & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (2017).